NOTATION · CHEAT SHEET

Limit Notation at a Glance

The marks from the notation cards across the limit pages: how each is read and what it means. Every group links to the full card on its page.

TWO-SIDED LIMITS

$x \to a$	**X APPROACHES A** x gets close to a without being a.
$\lim_{x \to a} f(x) = L$	**THE LIMIT OF F OF X AS X APPROACHES A IS L** Values of f settle at L near a; $f(a)$ itself plays no part.
$\lim_{x \to a} f(x)$ DNE	**THE LIMIT DOES NOT EXIST** No single value is approached, for example when the two sides disagree.

ONE-SIDED LIMITS

$\lim_{x \to a^-} f(x)$	**LIMIT FROM THE LEFT** Only $x < a$ counts; the minus is a direction, not a sign.
$\lim_{x \to a^+} f(x)$	**LIMIT FROM THE RIGHT** Only $x > a$ counts.
$f(a^-),\ f(a^+)$	**F OF A MINUS, F OF A PLUS** Short forms of the two one-sided limits.

LIMITS AT INFINITY

∞	**INFINITY** Not a number: a direction of unbounded growth.
$x \to \infty$	**X APPROACHES INFINITY** x grows without bound.
$\lim_{x \to a} f(x) = \infty$	**THE LIMIT IS INFINITY** f grows without bound near a; strictly, the limit does not exist.

EVALUATING LIMITS

$\frac{0}{0},\ \frac{\infty}{\infty}$	**ZERO OVER ZERO; INFINITY OVER INFINITY** Indeterminate: the form alone does not give the limit.
$0 \cdot \infty,\ 1^\infty,\ 0^0$	**PRODUCT AND POWER FORMS** Also indeterminate, with $\infty - \infty$ and ∞^0.
$\frac{1}{0},\ \frac{0}{\infty}$	**THE DETERMINATE LOOKALIKES** These do decide the answer: unbounded, and 0.
$\lim \cdots = \lim \cdots = L$	**THE EVALUATION CHAIN** Each equals sign carries the $\lim$ until the last step.

CONTINUITY

$\lim_{x \to a} f(x) = f(a)$	**THE LIMIT AT A EQUALS F OF A** The definition of continuity at a.
$f(a^+) = f(a)$	**F IS RIGHT-CONTINUOUS AT A** The right-hand limit equals the value: continuity from one side, as at an endpoint.
$f \in C[a, b]$	**F IS IN C OF A, B** The set of functions continuous on $[a, b]$.

SAME MARKS, DIFFERENT JOB

a^- vs $-a$	A superscript minus is a direction (from the left), not a negative number.
$= \infty$	Describes how the limit fails to exist; it is not a value.
$\frac{1}{0}$ vs $\frac{0}{0}$	The first is unbounded, the second tells nothing on its own.